



**UNIVERSIDAD
DE GRANADA**



IMAG
INSTITUTO DE MATEMÁTICAS
Universidad de Granada

Hermitian duality of dihedral codes and application to quantum codes

Víctor Sotomayor

Universidad de Granada

XIV International Workshop on Coding and Cryptography

Paris, June 2026

Joint work with Miguel Sales and Xaro Soler

Universitat d'Alacant (Spain)



GROUP CODES AND GROUP ALGEBRAS

Preliminaries

Notation

- G is a finite group
- $\mathbb{F}_q$ is the field of q elements (q a prime power)

Definition. The **group algebra** $\mathbb{F}_q[G]$ is the $\mathbb{F}_q$ -vector space

$$\mathbb{F}_q[G] = \left\{ \sum_{g \in G} \alpha_g g \ : \ \alpha_g \in \mathbb{F}_q \right\}$$

with basis the elements of G , and equipped with

$$\left(\sum_{g \in G} \alpha_g g \right) \cdot \left(\sum_{g \in G} \beta_g g \right) = \sum_{g \in G} \left(\sum_{h \in G} \alpha_h \beta_{h^{-1}g} \right) g$$

Remark. $(\mathbb{F}_q[G], +, \cdot)$ is a **ring**. Moreover, As $\mathbb{F}_q$ -vector space, it holds $\mathbb{F}_q[G] \cong \mathbb{F}_q^{|G|}$.



Preliminaries

Notation

- G is a finite group
- $\mathbb{F}_q$ is the field of q elements (q a prime power)

Definition. The **group algebra** $\mathbb{F}_q[G]$ is the $\mathbb{F}_q$ -vector space

$$\mathbb{F}_q[G] = \left\{ \sum_{g \in G} \alpha_g g \ : \ \alpha_g \in \mathbb{F}_q \right\}$$

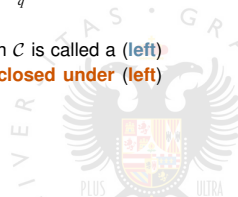
with basis the elements of G , and equipped with

$$\left(\sum_{g \in G} \alpha_g g \right) \cdot \left(\sum_{g \in G} \beta_g g \right) = \sum_{g \in G} \left(\sum_{h \in G} \alpha_h \beta_{h^{-1}g} \right) g$$

Remark. $(\mathbb{F}_q[G], +, \cdot)$ is a **ring**. Moreover, As $\mathbb{F}_q$ -vector space, it holds $\mathbb{F}_q[G] \cong \mathbb{F}_q^{|G|}$.

Definition. Let $\mathcal{C} \subseteq \mathbb{F}_q^n$ be a linear code, and let G be a group of order n . Then $\mathcal{C}$ is called a **(left) G -code** if there exists an isomorphism $\phi : \mathbb{F}_q^n \rightarrow \mathbb{F}_q[G]$ such that $\phi(\mathcal{C})$ is **closed under (left) multiplication** of every $g \in G$.

In other words, a **(left) G -code** is any (left) ideal of $\mathbb{F}_q[G]$.



Group codes

Remark. There is a bijection between left and right ideals of $\mathbb{F}_q[G]$, so we **only study left ideals**.

Example. Consider the linear code $\mathcal{C} = \{000000, 111111\} \subseteq \mathbb{F}_2^6$. Then $\mathcal{C}$ is a Σ_3 -code, since we can take the isomorphism $\phi : \mathbb{F}_2^6 \longrightarrow \mathbb{F}_2[\Sigma_3]$ such that

$$a_1 a_2 a_3 a_4 a_5 a_6 \longmapsto a_1 () + a_2 (1, 2) + a_3 (1, 3) + a_4 (2, 3) + a_5 (1, 2, 3) + a_6 (1, 3, 2),$$

and $\phi(\mathcal{C}) = \{\mathbf{0}, () + (1, 2) + (1, 3) + (2, 3) + (1, 2, 3) + (1, 3, 2)\}$ is certainly an ideal.

Observe that $\mathcal{C}$ is also an **abelian group code**, since it is a C_6 -code.



Group codes

Remark. There is a bijection between left and right ideals of $\mathbb{F}_q[G]$, so we **only study left ideals**.

Example. Consider the linear code $\mathcal{C} = \{000000, 111111\} \subseteq \mathbb{F}_2^6$. Then $\mathcal{C}$ is a Σ_3 -code, since we can take the isomorphism $\phi : \mathbb{F}_2^6 \longrightarrow \mathbb{F}_2[\Sigma_3]$ such that

$$a_1 a_2 a_3 a_4 a_5 a_6 \longmapsto a_1 () + a_2 (1, 2) + a_3 (1, 3) + a_4 (2, 3) + a_5 (1, 2, 3) + a_6 (1, 3, 2),$$

and $\phi(\mathcal{C}) = \{0, () + (1, 2) + (1, 3) + (2, 3) + (1, 2, 3) + (1, 3, 2)\}$ is certainly an ideal.

Observe that $\mathcal{C}$ is also an **abelian group code**, since it is a C_6 -code.

Notes.

- ▶ Group codes **generalise cyclic codes**: any cyclic code of length n is a C_n -code.
- ▶ Reed-Solomon and Reed-Muller codes are **abelian group codes** (G -codes for some abelian G).



Group codes

Remark. There is a bijection between left and right ideals of $\mathbb{F}_q[G]$, so we **only study left ideals**.

Example. Consider the linear code $\mathcal{C} = \{000000, 111111\} \subseteq \mathbb{F}_2^6$. Then $\mathcal{C}$ is a Σ_3 -code, since we can take the isomorphism $\phi : \mathbb{F}_2^6 \longrightarrow \mathbb{F}_2[\Sigma_3]$ such that

$$a_1 a_2 a_3 a_4 a_5 a_6 \longmapsto a_1 () + a_2 (1, 2) + a_3 (1, 3) + a_4 (2, 3) + a_5 (1, 2, 3) + a_6 (1, 3, 2),$$

and $\phi(\mathcal{C}) = \{0, () + (1, 2) + (1, 3) + (2, 3) + (1, 2, 3) + (1, 3, 2)\}$ is certainly an ideal.

Observe that $\mathcal{C}$ is also an **abelian group code**, since it is a C_6 -code.

Notes.

- ▶ Group codes **generalise cyclic codes**: any cyclic code of length n is a C_n -code.
- ▶ Reed-Solomon and Reed-Muller codes are **abelian group codes** (G -codes for some abelian G).

Remark. There are linear codes that **cannot be realised** as G -codes for some group G .

Theorem (Bernal, del Río, Simón - 2009).

A linear code $\mathcal{C}$ is a G -code if and only if $G \lesssim \text{PAut}(\mathcal{C})$ and G is transitive on $\{1, \dots, n\}$.



Semisimple group algebras

Representation theory is a powerful tool to study group codes, i.e. ideals in $\mathbb{F}_q[G]$.

Theorem (Maschke + Wedderburn-Artin). If q does not divide $|G|$, then $\mathbb{F}_q[G]$ is semisimple, and

$$\mathbb{F}_q[G] \cong \bigoplus_{i=1}^s \mathcal{M}_{n_i}(\mathbb{F}_q r_i)$$

as $\mathbb{F}_q$ -algebras, where $|G| = n_1^2 r_1 + \cdots + n_s^2 r_s$.

Important consequences.

► **Every ideal** $I \subseteq \mathbb{F}_q[G]$ verifies $I \cong \langle M_1 \rangle \oplus \cdots \oplus \langle M_s \rangle$ **with M_i in RREF**.

► Whenever the Wedderburn-Artin's decomposition of $\mathbb{F}_q[G]$ is known, we can easily:

- **enumerate** all possible G -codes;
- **compute the dimension** over $\mathbb{F}_q$ of any G -code, which is $\sum_{i=1}^s n_i r_i \text{rank}(M_i)$.

► The minimum **distance satisfies** $d(\mathcal{C}) \geq \frac{|G|}{\dim(\mathcal{C})}$ (Borello, Willems, Zini - 2022).



Some group algebra decompositions

- Let C_n be a **cyclic group** of order n , with $(q, n) = 1$.

If $x^n - 1 = f_1 \cdots f_r \in \mathbb{F}_q[x]$ with f_j irreducible, then applying Chinese Remainder Theorem:

$$\mathbb{F}_q[C_n] \cong \bigoplus_{i=1}^r \mathbb{F}_{q^{\deg(f_j)}}$$

- Let G be an **abelian group** of order n , with $(q, n) = 1$. For each $d \mid n$, set

$$\begin{cases} \alpha_d \text{ a primitive } d\text{-th root of unity} \\ t_d = |\mathbb{F}_q(\alpha_d) : \mathbb{F}_q| \\ n_d \text{ the number of elements of order } d \text{ in } G \\ a_d = \frac{n_d}{t_d} \end{cases}$$

In virtue of the so-called Perlis-Walker Theorem (1950):

$$\mathbb{F}_q[G] \cong \bigoplus_{d \mid n} \left(\mathbb{F}_{q^{t_d}} \right)^{\oplus a_d}$$

But **non-abelian** group algebras have a **richer algebraic structure**, that may provide

$$\begin{cases} \text{codes with } \textbf{good parameters} \\ \textbf{post-quantum} \text{ public-key encryption protocols} \end{cases}$$



The dihedral group algebra decomposition

- Let D_n be a **dihedral group** of order $2n$, with **$(q, 2n) = 1$** .

Let $x^n - 1 = f_1 \cdots f_r f_{r+1} f_{r+1}^* \cdots f_{r+s} f_{r+s}^* \in \mathbb{F}_q[x]$ with f_j monic, irreducible, and

$$f_j^*(x) = f_j(0)^{-1} x^{\deg(f_j)} \cdot f_j(x^{-1}) \quad (\text{reciprocal polynomials})$$

satisfying $f_1 = x - 1, f_2 = x + 1$ if n is even, and $f_j = f_j^*$, for every $1 \leq j \leq r$.

Set $\zeta(n) = 1$ if n is odd, and $\zeta(n) = 2$ if n is even. Brochero-Martínez proved in 2015:

$$\mathbb{F}_q[D_n] \cong \bigoplus_{j=1}^{r+s} A_j, \text{ where } A_j = \begin{cases} \mathbb{F}_q \oplus \mathbb{F}_q, & \text{if } 1 \leq j \leq \zeta(n) \\ \mathcal{M}_2(\mathbb{F}_q(\alpha_j + \alpha_j^{-1})), & \text{if } \zeta(n) + 1 \leq j \leq r \\ \mathcal{M}_2(\mathbb{F}_q(\alpha_j)), & \text{if } r + 1 \leq j \leq r + s \end{cases},$$

where α_j is a root of f_j . Note $\mathbb{F}_q(\alpha_j) \cong \mathbb{F}_{q^{\deg(f_j)}}$ and $\mathbb{F}_q(\alpha_j + \alpha_j^{-1}) \cong \mathbb{F}_{q^{\deg(f_j)/2}}$.



The dihedral group algebra decomposition

- Let D_n be a **dihedral group** of order $2n$, with $(q, 2n) = 1$.

Let $x^n - 1 = f_1 \cdots f_r f_{r+1} f_{r+1}^* \cdots f_{r+s} f_{r+s}^* \in \mathbb{F}_q[x]$ with f_j monic, irreducible, and

$$f_j^*(x) = f_j(0)^{-1} x^{\deg(f_j)} \cdot f_j(x^{-1}) \quad (\text{reciprocal polynomials})$$

satisfying $f_1 = x - 1, f_2 = x + 1$ if n is even, and $f_j = f_j^*$, for every $1 \leq j \leq r$.

Set $\zeta(n) = 1$ if n is odd, and $\zeta(n) = 2$ if n is even. Brochero-Martínez proved in 2015:

$$\mathbb{F}_q[D_n] \cong \bigoplus_{j=1}^{r+s} A_j, \text{ where } A_j = \begin{cases} \mathbb{F}_q \oplus \mathbb{F}_q, & \text{if } 1 \leq j \leq \zeta(n) \\ \mathcal{M}_2(\mathbb{F}_q(\alpha_j + \alpha_j^{-1})), & \text{if } \zeta(n) + 1 \leq j \leq r \\ \mathcal{M}_2(\mathbb{F}_q(\alpha_j)), & \text{if } r + 1 \leq j \leq r + s \end{cases},$$

where α_j is a root of f_j . Note $\mathbb{F}_q(\alpha_j) \cong \mathbb{F}_{q^{\deg(f_j)}}$ and $\mathbb{F}_q(\alpha_j + \alpha_j^{-1}) \cong \mathbb{F}_{q^{\deg(f_j)/2}}$.

Example. Let $q = 7$ and $n = 15$, so $G = D_{15}$ is a **dihedral group** of order 30. Then

$$\begin{aligned} x^{15} - 1 &= \underbrace{(x - 1)}_{f_1} \cdot \underbrace{(x^4 + x^3 + x^2 + x + 1)}_{f_2} \cdot \underbrace{(x + \xi)}_{f_3} \cdot \underbrace{(x + \xi^5)}_{f_3^*} \cdot \\ &\quad \cdot \underbrace{(x^4 + \xi^2 x^3 + \xi^4 x^2 + x + \xi^2)}_{f_4} \cdot \underbrace{(x^4 + \xi^4 x^3 + \xi^2 x^2 + x + \xi^4)}_{f_4^*}, \end{aligned}$$

where $\xi \in \mathbb{F}_7^\times$ is a generator, so $r = 2$ and $s = 2$. Note $\zeta(n) = 1$. Hence

$$\mathbb{F}_7[D_{15}] \cong (\mathbb{F}_7 \oplus \mathbb{F}_7)_{A_1} \oplus \mathcal{M}_2(\mathbb{F}_7)_{A_2} \oplus \mathcal{M}_2(\mathbb{F}_7)_{A_3} \oplus \mathcal{M}_2(\mathbb{F}_7)_{A_4}.$$



The dihedral group algebra decomposition

- Let D_n be a **dihedral group** of order $2n$, with **$(q, 2n) = 1$** .

Let $x^n - 1 = f_1 \cdots f_r f_{r+1} f_{r+1}^* \cdots f_{r+s} f_{r+s}^* \in \mathbb{F}_q[x]$ with f_j monic, irreducible, and

$$f_j^*(x) = f_j(0)^{-1} x^{\deg(f_j)} \cdot f_j(x^{-1}) \quad (\text{reciprocal polynomials})$$

satisfying $f_1 = x - 1$, $f_2 = x + 1$ if n is even, and $f_j = f_j^*$, for every $1 \leq j \leq r$.

Set $\zeta(n) = 1$ if n is odd, and $\zeta(n) = 2$ if n is even. Brochero-Martínez proved in 2015:

$$\mathbb{F}_q[D_n] \cong \bigoplus_{j=1}^{r+s} A_j, \text{ where } A_j = \begin{cases} \mathbb{F}_q \oplus \mathbb{F}_q, & \text{if } 1 \leq j \leq \zeta(n) \\ \mathcal{M}_2(\mathbb{F}_q(\alpha_j + \alpha_j^{-1})), & \text{if } \zeta(n) + 1 \leq j \leq r \\ \mathcal{M}_2(\mathbb{F}_q(\alpha_j)), & \text{if } r + 1 \leq j \leq r + s \end{cases},$$

where α_j is a root of f_j . Note $\mathbb{F}_q(\alpha_j) \cong \mathbb{F}_{q^{\deg(f_j)}}$ and $\mathbb{F}_q(\alpha_j + \alpha_j^{-1}) \cong \mathbb{F}_{q^{\deg(f_j)/2}}$.

Remark. When $q = 2^m$ but n is odd, then the above isomorphism also works by simply replacing

$$A_1 = \mathbb{F}_q[C_2],$$

where C_2 denotes the cyclic group generated by $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

Hence we can consider both the semisimple case and this last situation, allowing in particular the construction of **binary codes**.



ON DUALITIES OF DIHEDRAL CODES

On dualities of group codes

Remark. If the group algebra is decomposed as $\mathbb{F}_q[G] \cong \bigoplus_{i=1}^s \mathcal{M}_{n_i}(\mathbb{F}_{q^{r_i}})$, and

$$I \cong I_1 \oplus \cdots \oplus I_s$$

with I_i an ideal of $\mathcal{M}_{n_i}(\mathbb{F}_{q^{r_i}})$ for every $1 \leq i \leq s$, then

$$I^{\perp_e} \cong I_1^{\perp_e} \oplus \cdots \oplus I_s^{\perp_e} \quad \text{and} \quad I^{\perp_h} \cong I_1^{\perp_h} \oplus \cdots \oplus I_s^{\perp_h},$$

where the euclidean (hermitian) metric in $\mathbb{F}_q[G]$ ($\mathbb{F}_{q^2}[G]$) is the natural extension of that in $\mathbb{F}_q^n$ ($\mathbb{F}_{q^2}^n$).

Hence, it is enough to find the dual (either euclidean or hermitian) of each I_i in $\mathcal{M}_{n_i}(\mathbb{F}_{q^{r_i}})$.



On dualities of group codes

Remark. If the group algebra is decomposed as $\mathbb{F}_q[G] \cong \bigoplus_{i=1}^s \mathcal{M}_{n_i}(\mathbb{F}_q r_i)$, and

$$I \cong I_1 \oplus \cdots \oplus I_s$$

with I_i an ideal of $\mathcal{M}_{n_i}(\mathbb{F}_q r_i)$ for every $1 \leq i \leq s$, then

$$I^{\perp_e} \cong I_1^{\perp_e} \oplus \cdots \oplus I_s^{\perp_e} \quad \text{and} \quad I^{\perp_h} \cong I_1^{\perp_h} \oplus \cdots \oplus I_s^{\perp_h},$$

where the euclidean (hermitian) metric in $\mathbb{F}_q[G]$ ($\mathbb{F}_{q^2}[G]$) is the natural extension of that in $\mathbb{F}_q^n$ ($\mathbb{F}_{q^2}^n$).

Hence, it is enough to find the dual (either euclidean or hermitian) of each I_i in $\mathcal{M}_{n_i}(\mathbb{F}_q r_i)$.

The **euclidean dual** of a (left) **dihedral code** in $\mathbb{F}_q[D_n]$ was completely described by Vedenov and Deundyak (2021) when $(q, n) = 1$.

Question. Given a dihedral code $I \subseteq \mathbb{F}_{q^2}[D_n]$ with $(q, n) = 1$,

can we obtain a complete description of its hermitian dual code $I^{\perp_h}$?



A reformulation of the decomposition of $\mathbb{F}_{q^2}[D_n]$

Given $f = \sum_{i=0}^m a_i x^i \in \mathbb{F}_{q^2}[x]$, its **conjugate polynomial** is $\bar{f} = \sum_{i=0}^m a_i^q x^i$.

If $(q, n) = 1$, then we can factorise $x^n - 1 \in \mathbb{F}_{q^2}[x]$ as follows:

$$x^n - 1 = \prod_{j \in J_0} f_j \prod_{j \in J_1} f_j \bar{f}_j \prod_{j \in J_2 \cup J_3} f_j f_j^* \prod_{j \in J_4} f_j f_j^* \bar{f}_j \bar{f}_j^*,$$

where:

- ▶ $j \in J_0 \iff \bar{f}_j = f_j = f_j^*$. In this case, $f_j \in \{x - 1, x + 1\}$
- ▶ $j \in J_1 \iff \bar{f}_j \neq f_j = f_j^*$
- ▶ $j \in J_2 \iff \bar{f}_j = f_j \neq f_j^*$
- ▶ $j \in J_3 \iff \bar{f}_j = f_j^* \neq f_j$
- ▶ $j \in J_4 \iff f_j, \bar{f}_j \text{ and } f_j^* \text{ are all different}$



A reformulation of the decomposition of $\mathbb{F}_{q^2}[D_n]$

Given $f = \sum_{i=0}^m a_i x^i \in \mathbb{F}_{q^2}[x]$, its **conjugate polynomial** is $\bar{f} = \sum_{i=0}^m a_i^q x^i$.

If $(q, n) = 1$, then we can factorise $x^n - 1 \in \mathbb{F}_{q^2}[x]$ as follows:

$$x^n - 1 = \prod_{j \in J_0} f_j \prod_{j \in J_1} f_j \bar{f}_j \prod_{j \in J_2 \cup J_3} f_j f_j^* \prod_{j \in J_4} f_j f_j^* \bar{f}_j \bar{f}_j^*,$$

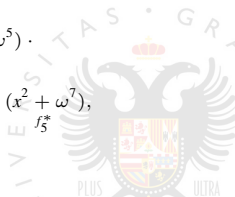
where:

- ▶ $j \in J_0 \iff \bar{f}_j = f_j = f_j^*$. In this case, $f_j \in \{x - 1, x + 1\}$
- ▶ $j \in J_1 \iff \bar{f}_j \neq f_j = f_j^*$
- ▶ $j \in J_2 \iff \bar{f}_j = f_j \neq f_j^*$
- ▶ $j \in J_3 \iff \bar{f}_j = f_j^* \neq f_j$
- ▶ $j \in J_4 \iff f_j, \bar{f}_j \text{ and } f_j^* \text{ are all different}$

Example. Let $q = 3$ and $n = 16$. Take a generator ω of $\mathbb{F}_9^\times$. Then in $\mathbb{F}_9[x]$:

$$\begin{aligned} x^{16} - 1 &= \underbrace{(x - 1)}_{f_1} \cdot \underbrace{(x + 1)}_{f_2} \cdot \underbrace{(x + \omega)}_{f_3} \cdot \underbrace{(x + \omega^2)}_{f_4} \cdot \underbrace{(x + \omega^3)}_{\bar{f}_3} \cdot \underbrace{(x + \omega^5)}_{\bar{f}_3^*} \cdot \\ &\quad \cdot \underbrace{(x + \omega^6)}_{\bar{f}_4 = f_4^*} \cdot \underbrace{(x + \omega^7)}_{f_3^*} \cdot \underbrace{(x^2 + \omega)}_{f_5} \cdot \underbrace{(x^2 + \omega^3)}_{\bar{f}_5} \cdot \underbrace{(x^2 + \omega^5)}_{\bar{f}_5^*} \cdot \underbrace{(x^2 + \omega^7)}_{f_5^*}, \end{aligned}$$

so $J_0 = \{1, 2\}$, $J_1 = J_2 = \emptyset$, $J_3 = \{4\}$ and $J_4 = \{3, 5\}$.



A reformulation of the decomposition of $\mathbb{F}_{q^2}[D_n]$

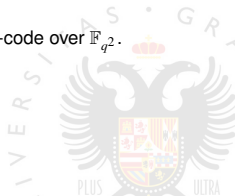
For $(q, n) = 1$, consider the **previous decomposition** of $x^n - 1$ in $\mathbb{F}_{q^2}[x]$. Take α_j a **root** of f_j .

Theorem. There exists an isomorphism of $\mathbb{F}_{q^2}$ -algebras $\rho : \mathbb{F}_{q^2}[D_n] \longrightarrow \bigoplus_j A_j$ with

$$A_j = \begin{cases} \mathbb{F}_{q^2} \oplus \mathbb{F}_{q^2} & \text{if } j \in J_0 \text{ and } q \text{ odd} \\ \mathbb{F}_{q^2}[C_2] & \text{if } j \in J_0 \text{ and } q = 2^m \\ \mathcal{M}_2\left(\mathbb{F}_{q^2}(\alpha_j + \alpha_j^{-1})\right) \oplus \mathcal{M}_2\left(\mathbb{F}_{q^2}(\alpha_j^q + \alpha_j^{-q})\right) & \text{if } j \in J_1 \\ \mathcal{M}_2\left(\mathbb{F}_{q^2}(\alpha_j)\right) & \text{if } j \in J_2 \cup J_3 \\ \mathcal{M}_2\left(\mathbb{F}_{q^2}(\alpha_j)\right) \oplus \mathcal{M}_2\left(\mathbb{F}_{q^2}(\alpha_j^q)\right) & \text{if } j \in J_4 \end{cases}.$$

Note $\mathbb{F}_{q^2}(\alpha_j) \cong \mathbb{F}_{q^{2 \deg(f_j)}} \cong \mathbb{F}_{q^2}(\alpha_j^q)$ and $\mathbb{F}_{q^2}(\alpha_j + \alpha_j^{-1}) \cong \mathbb{F}_{q^{\deg(f_j)}} \cong \mathbb{F}_{q^2}(\alpha_j^q + \alpha_j^{-q})$.

► We used the **isomorphism** ρ to fully describe the **hermitian dual** of any D_n -code over $\mathbb{F}_{q^2}$.



The hermitian dual of a D_n -code

Assume $(q, n) = 1$, and let $\mathcal{C}$ be a D_n -code over $\mathbb{F}_{q^2}$. Set $J = J_0 \cup J_1 \cup J_2 \cup J_3 \cup J_4$.

Theorem. If $\mathcal{C} \cong \bigoplus_{j \in J} I_j \subseteq \bigoplus_{j \in J} A_j$, then $\mathcal{C}^{\perp_h} \cong \bigoplus_{j \in J} I_j^{\perp_h} \subseteq \bigoplus_{j \in J} A_j$, where:

(i) for $j \in J_0$ and q odd:

$$I_j^{\perp_h} = \begin{cases} \mathbb{F}_{q^2} \oplus \mathbb{F}_{q^2} & \text{if } I_j = \mathbf{0} \oplus \mathbf{0} \\ \mathbf{0} \oplus \mathbb{F}_{q^2} & \text{if } I_j = \mathbb{F}_{q^2} \oplus \mathbf{0} \\ \mathbb{F}_{q^2} \oplus \mathbf{0} & \text{if } I_j = \mathbf{0} \oplus \mathbb{F}_{q^2} \\ \mathbf{0} \oplus \mathbf{0} & \text{if } I_j = \mathbb{F}_{q^2} \oplus \mathbb{F}_{q^2} \end{cases}$$

(ii) for $j \in J_0$ and $q = 2^m$:

$$I_j^{\perp_h} = \begin{cases} \mathbb{F}_{q^2}[C_2] & \text{if } I_j = \mathbf{0} \\ \left\langle \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right\rangle & \text{if } I_j = \left\langle \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right\rangle \\ \mathbf{0} & \text{if } I_j = \mathbb{F}_{q^2}[C_2] \end{cases}$$



The hermitian dual of a D_n -code

Assume $(q, n) = 1$, and let $\mathcal{C}$ be a D_n -code over $\mathbb{F}_{q^2}$. Set $J = J_0 \cup J_1 \cup J_2 \cup J_3 \cup J_4$.

Theorem. If $\mathcal{C} \cong \bigoplus_{j \in J} I_j \subseteq \bigoplus_{j \in J} A_j$, then $\mathcal{C}^{\perp_h} \cong \bigoplus_{j \in J} I_j^{\perp_h} \subseteq \bigoplus_{j \in J} A_j$, where:

(iv) for $j \in J_2$:

$$I_j^{\perp_h} = \begin{cases} A_j & \text{if } I_j = \mathbf{0} \\ \mathbf{0} & \text{if } I_j = A_j \\ \left\langle \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle & \text{if } I_j = \left\langle \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle \\ \left\langle \begin{bmatrix} 1 & -\lambda_j^q \deg(f_j) \\ 0 & 0 \end{bmatrix} \right\rangle & \text{if } I_j = \left\langle \begin{bmatrix} 1 & \lambda_j \\ 0 & 0 \end{bmatrix} \right\rangle, \lambda_j \in \mathbb{F}_{q^2}(\alpha_j) \cong \mathbb{F}_{q^2}^{\deg(f_j)} \end{cases}$$



The hermitian dual of a D_n -code

Assume $(q, n) = 1$, and let $\mathcal{C}$ be a D_n -code over $\mathbb{F}_{q^2}$. Set $J = J_0 \cup J_1 \cup J_2 \cup J_3 \cup J_4$.

Theorem. If $\mathcal{C} \cong \bigoplus_{j \in J} I_j \subseteq \bigoplus_{j \in J} A_j$, then $\mathcal{C}^{\perp_h} \cong \bigoplus_{j \in J} I_j^{\perp_h} \subseteq \bigoplus_{j \in J} A_j$, where:

(v) for $j \in J_3$:

$$I_j^{\perp_h} = \begin{cases} A_j & \text{if } I_j = \mathbf{0} \\ \mathbf{0} & \text{if } I_j = A_j \\ \left\langle \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\rangle & \text{if } I_j = \left\langle \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle \\ \left\langle \begin{bmatrix} -\lambda_j^{q^{\deg(f_j)}} & 1 \\ 0 & 0 \end{bmatrix} \right\rangle & \text{if } I_j = \left\langle \begin{bmatrix} 1 & \lambda_j \\ 0 & 0 \end{bmatrix} \right\rangle, \lambda_j \in \mathbb{F}_{q^2}(\alpha_j) \cong \mathbb{F}_{q^{2 \deg(f_j)}} \end{cases}$$

For $j \in J_1 \cup J_4$ the expression of $I_j^{\perp_h}$ is a bit more technique...



The hermitian dual of a D_n -code

Example (continue). Let $q = 3$ and $n = 16$, so $J_0 = \{1, 2\}$, $J_1 = J_2 = \emptyset$, $J_3 = \{4\}$ and $J_4 = \{3, 5\}$, with $\deg(f_3) = 1$, $\deg(f_5) = 2$ and $\deg(f_4) = 1$. We have

$$\mathbb{F}_9[D_{16}] \stackrel{\rho}{\cong} (\mathbb{F}_9 \oplus \mathbb{F}_9)_{A_1} \oplus (\mathbb{F}_9 \oplus \mathbb{F}_9)_{A_2} \oplus \mathcal{M}_2(\mathbb{F}_9)_{A_4} \oplus (\mathcal{M}_2(\mathbb{F}_9) \oplus \mathcal{M}_2(\mathbb{F}_9))_{A_3} \oplus (\mathcal{M}_2(\mathbb{F}_{9^2}) \oplus \mathcal{M}_2(\mathbb{F}_{9^2}))_{A_5}.$$

Set $\mathbb{F}_9^\times = \langle \omega \rangle$ and $\mathbb{F}_{9^2}^\times = \langle \nu \rangle$. Consider as $\mathcal{C}$ the following **D_{16} -code** over $\mathbb{F}_9$:

$$\mathcal{C} \cong (\mathbf{0} \oplus \mathbf{0})_{I_1} \oplus (\mathbf{0} \oplus \mathbf{0})_{I_2} \oplus \left\langle \begin{bmatrix} 1 & \omega^7 \\ 0 & 0 \end{bmatrix} \right\rangle_{I_4} \oplus \left\langle \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right\rangle_{I_3} \oplus \left\langle \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 1 & \nu^{14} \\ 0 & 0 \end{bmatrix} \right\rangle_{I_5} \oplus \left\langle \begin{bmatrix} 1 & \nu^2 \\ 0 & 0 \end{bmatrix} \right\rangle.$$

Applying our main theorem, we obtained

$$\mathcal{C}^{\perp_h} \cong (\mathbb{F}_9 \oplus \mathbb{F}_9)_{I_1^{\perp_h}} \oplus (\mathbb{F}_9 \oplus \mathbb{F}_9)_{I_2^{\perp_h}} \oplus \left\langle \begin{bmatrix} 1 & \omega^7 \\ 0 & 0 \end{bmatrix} \right\rangle_{I_4^{\perp_h}} \oplus \left\langle \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \right\rangle_{I_3^{\perp_h}} \oplus \mathcal{M}_2(\mathbb{F}_9) \oplus \left\langle \begin{bmatrix} 1 & \nu^{14} \\ 0 & 0 \end{bmatrix} \right\rangle_{I_5^{\perp_h}} \oplus \left\langle \begin{bmatrix} 1 & \nu^2 \\ 0 & 0 \end{bmatrix} \right\rangle.$$



The hermitian dual of a D_n -code

Example (continue). Let $q = 3$ and $n = 16$, so $J_0 = \{1, 2\}$, $J_1 = J_2 = \emptyset$, $J_3 = \{4\}$ and $J_4 = \{3, 5\}$, with $\deg(f_3) = 1$, $\deg(f_5) = 2$ and $\deg(f_4) = 1$. We have

$$\mathbb{F}_9[D_{16}] \stackrel{p}{\cong} (\mathbb{F}_9 \oplus \mathbb{F}_9) \oplus (\mathbb{F}_9 \oplus \mathbb{F}_9) \oplus \mathcal{M}_2(\mathbb{F}_9) \oplus (\mathcal{M}_2(\mathbb{F}_9) \oplus \mathcal{M}_2(\mathbb{F}_9)) \oplus (\mathcal{M}_2(\mathbb{F}_{9^2}) \oplus \mathcal{M}_2(\mathbb{F}_{9^2})).$$

$A_1 \quad A_2 \quad A_4 \quad A_3 \quad A_5$

Set $\mathbb{F}_9^\times = \langle \omega \rangle$ and $\mathbb{F}_{9^2}^\times = \langle \nu \rangle$. Consider as $\mathcal{C}$ the following **D_{16} -code** over $\mathbb{F}_9$:

$$\mathcal{C} \cong (\mathbf{0} \oplus \mathbf{0}) \oplus (\mathbf{0} \oplus \mathbf{0}) \oplus \left\langle \begin{bmatrix} 1 & \omega^7 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 1 & \nu^{14} \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 1 & \nu^2 \\ 0 & 0 \end{bmatrix} \right\rangle.$$

$I_1 \quad I_2 \quad I_4 \quad I_3 \quad I_5$

Applying our main theorem, we obtained

$$\mathcal{C}^{\perp_h} \cong (\mathbb{F}_9 \oplus \mathbb{F}_9) \oplus (\mathbb{F}_9 \oplus \mathbb{F}_9) \oplus \left\langle \begin{bmatrix} 1 & \omega^7 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \mathcal{M}_2(\mathbb{F}_9) \oplus \left\langle \begin{bmatrix} 1 & \nu^{14} \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 1 & \nu^2 \\ 0 & 0 \end{bmatrix} \right\rangle.$$

$I_1^{\perp_h} \quad I_2^{\perp_h} \quad I_4^{\perp_h} \quad I_3^{\perp_h} \quad I_5^{\perp_h}$

Remark. Note that $\mathcal{C} \subseteq \mathcal{C}^{\perp_h}$, so $\mathcal{C}$ is an **hermitian self-orthogonal code** over $\mathbb{F}_9$.



LCD and SO dihedral codes

Our algebraic description of the hermitian dual of a D_n -code is useful in several code constructions:

- ▶ For any $I \subseteq \mathbb{F}_{q^2}[D_n]$, it is easy to compute the **hermitian hull** $I \cap I^{\perp_h}$, and check if $I \cap I^{\perp_h} = \mathbf{0}$, so we can produce **all** distinct hermitian **Linear Complementary Dual** D_n -codes over $\mathbb{F}_{q^2}$.
- ▶ One can construct **all** D_n -codes that are **hermitian self-orthogonal** over $\mathbb{F}_{q^2}$.

Remark. Hermitian self-orthogonal codes are very useful to build **quantum error correcting codes**.



Hermitian self-orthogonal D_n -codes

Assume $(q, n) = 1$, and let $\mathcal{C} \subseteq \mathbb{F}_{q^2}[D_n]$ be a D_n -code. Set $J = J_0 \cup J_1 \cup J_2 \cup J_3 \cup J_4$.

Corollary. If $\mathcal{C} \cong \bigoplus_{j \in J} I_j \subseteq \bigoplus_{j \in J} A_j$, then $\mathcal{C} \subseteq \mathcal{C}^{\perp_h}$ if and only if the following conditions are satisfied:

- (i) If $j \in J_0$ and q is odd, then $I_j = \mathbf{0} \oplus \mathbf{0}$.
- (ii) If $j \in J_0$ and $q = 2^m$, then $I_j = \mathbf{0}$ or $I_j = \left\langle \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right\rangle$.
- (iii) If $j \in J_1$, then either one of the summands of I_j is $\mathbf{0}$, or one of the following options holds for I_j :
 - ▶ $\left\langle \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 2 & -(\alpha_j^q + \alpha_j^{-q}) \\ 0 & 0 \end{bmatrix} \right\rangle$; or
 - ▶ $\left\langle \begin{bmatrix} 1 & \lambda_j \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} (2\lambda_j + \alpha_j + \alpha_j^{-1})^q & (-2 - \lambda_j(\alpha_j + \alpha_j^{-1}))^q \\ 0 & 0 \end{bmatrix} \right\rangle$,

where α_j is a root of f_j , and $\lambda_j \in \mathbb{F}_{q^2}(\alpha_j + \alpha_j^{-1}) \cong \mathbb{F}_{q^{\deg(f_j)}}$.



Hermitian self-orthogonal D_n -codes

Assume $(q, n) = 1$, and let $\mathcal{C} \subseteq \mathbb{F}_{q^2}[D_n]$ be a D_n -code. Set $J = J_0 \cup J_1 \cup J_2 \cup J_3 \cup J_4$.

Corollary. If $\mathcal{C} \cong \bigoplus_{j \in J} I_j \subseteq \bigoplus_{j \in J} A_j$, then $\mathcal{C} \subseteq \mathcal{C}^{\perp_h}$ if and only if the following conditions are satisfied:

(iv) If $j \in J_2$, then I_j is either $\mathbf{0}$, $\langle \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \rangle$, or $\langle \begin{bmatrix} 1 & \lambda_j \\ 0 & 0 \end{bmatrix} \rangle$ with $\lambda_j = -\lambda_j^{q^j}$, $\lambda_j \in \mathbb{F}_{q^2}(\alpha_j) \cong \mathbb{F}_{q^{2 \deg(f_j)}}$.

(v) If $j \in J_3$, then I_j is either $\mathbf{0}$, or $\langle \begin{bmatrix} 1 & \lambda_j \\ 0 & 0 \end{bmatrix} \rangle$ with $0 \neq \lambda_j = -\lambda_j^{-q^{r_j}}$, $\lambda_j \in \mathbb{F}_{q^2}(\alpha_j) \cong \mathbb{F}_{q^{2 \deg(f_j)}}$.

(vi) If $j \in J_4$, then either one of the summands of I_j is $\mathbf{0}$, or one of the following options holds for I_j :

► $\langle \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \rangle \oplus \langle \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \rangle$; or

► $\langle \begin{bmatrix} 1 & \lambda_j \\ 0 & 0 \end{bmatrix} \rangle \oplus \langle \begin{bmatrix} 1 & -\lambda_j^q \\ 0 & 0 \end{bmatrix} \rangle$ with $\lambda_j \in \mathbb{F}_{q^2}(\alpha_j) \cong \mathbb{F}_{q^{2 \deg(f_j)}}$.

Corollary. If $(q, n) = 1$, then the number of hermitian self-orthogonal D_n -codes over $\mathbb{F}_{q^2}$ is

$$\epsilon \prod_{j \in J_1} (3q^{r_j} + 6) \prod_{j \in J_2 \cup J_3} (q^{r_j} + 2) \prod_{j \in J_4} (3q^{2r_j} + 6),$$

with $\epsilon = 1$ if q is odd, and $\epsilon = 2$ otherwise.



DIHEDRAL QUANTUM CODES

Quantum error correcting codes

Definition. A q -ary **quantum error-correcting code** (QECC) $\mathcal{Q}$ of length n is a subspace of

$$\mathbb{C}^q \otimes \dots \otimes_n \mathbb{C}^q.$$

In particular $\dim_{\mathbb{C}}(\mathcal{Q}) \leq q^n$. Here we focus on the so-called **stabiliser codes**. In this case:

- ▶ $\dim_{\mathbb{C}}(\mathcal{Q}) = q^k$ for some positive integer $k \leq n$.
- ▶ We write its parameters as $[[n, k, d]]_q$, where the minimum distance d means that any error acting on $\leq d - 1$ positions of the tensor factors (**qubits**) can be detected or has no effect.

Two ways to construct such codes:

- ▶ using two linear codes over $\mathbb{F}_q$ such that $\mathcal{C}_1 \subseteq \mathcal{C}_2^{\perp_e}$ [**CSS construction**]
- ▶ using a linear code over $\mathbb{F}_{q^2}$ such that $\mathcal{C} \subseteq \mathcal{C}^{\perp_h}$ [**hermitian construction** – usually better]

Theorem. If $\mathcal{C} \subseteq \mathbb{F}_{q^2}^n$ is a linear code with parameters $[n, k, d]$ such that $\mathcal{C} \subseteq \mathcal{C}^{\perp_h}$, then there exists a quantum stabiliser code $\mathcal{Q}$ with parameters $[[n, n - 2k, d_{\mathcal{Q}}]]_q$, where

$$d_{\mathcal{Q}} := \min\{\text{wgt}(x) \mid x \in \mathcal{C}^{\perp_h} \setminus \mathcal{C}\} \geq d,$$

or $d_{\mathcal{Q}} := d$ if $\mathcal{C} = \mathcal{C}^{\perp_h}$.



Dihedral quantum codes

Example (continue). Let $q = 3$ and $n = 16$. Recall that

$$\mathbb{F}_9[D_{16}] \cong (\mathbb{F}_9 \oplus \mathbb{F}_9) \oplus (\mathbb{F}_9 \oplus \mathbb{F}_9) \oplus \mathcal{M}_2(\mathbb{F}_9) \oplus (\mathcal{M}_2(\mathbb{F}_9) \oplus \mathcal{M}_2(\mathbb{F}_9)) \oplus (\mathcal{M}_2(\mathbb{F}_{9^2}) \oplus \mathcal{M}_2(\mathbb{F}_{9^2})).$$

$A_1 \qquad A_2 \qquad A_4 \qquad A_3 \qquad A_5$

Set $\mathbb{F}_9^\times = \langle \omega \rangle$ and $\mathbb{F}_{9^2}^\times = \langle \nu \rangle$. Take the next **D_{16} -code** over $\mathbb{F}_9$:

$$\mathcal{C} \cong (\mathbf{0} \oplus \mathbf{0}) \oplus (\mathbf{0} \oplus \mathbf{0}) \oplus \left\langle \begin{bmatrix} 1 & \omega^7 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 1 & \nu^{14} \\ 0 & 0 \end{bmatrix} \right\rangle \oplus \left\langle \begin{bmatrix} 1 & \nu^2 \\ 0 & 0 \end{bmatrix} \right\rangle.$$

$I_1 \qquad I_2 \qquad I_4 \qquad I_3 \qquad I_5$

- We saw that $\mathcal{C}$ is an **hermitian self-orthogonal** code over $\mathbb{F}_9$. Its parameters are $[32, 12, 12]$.
- Applying the **hermitian construction**, $\exists \mathcal{Q}$ a quantum D_{16} -code with parameters $[[32, 8, 8]]_3$.
- $\mathcal{Q}$ reaches the **best known parameters** (for its length and dimension).





Merci de votre attention

Thanks for your attention

